prove a theorem that guarantees that the conformal map from the upper half-plane to a simply connected region bounded by a polygonal line is essentially given by a Schwarz-Christoffel integral.

4.2 The Schwarz-Christoffel integral

With the examples of the previous section in mind, we define the general **Schwarz-Christoffel integral** by

$$S(z) = \int_0^z \frac{d\zeta}{(\zeta - A_1)^{\beta_1} \cdots (\zeta - A_n)^{\beta_n}}. \tag{5}$$

Here $A_1 < A_2 < \cdots < A_n$ are n distinct points on the real axis arranged in increasing order. The exponents β_k will be assumed to satisfy the conditions $\beta_k < 1$ for each k and $1 < \sum_{k=1}^n \beta_k$.[6]

The integrand in (5) is defined as follows: $(z - A_k)^{\beta_k}$ is that branch (defined in the complex plane slit along the infinite ray $\{A_k + iy : y \leq 0\}$) which is positive when $z = x$ is real and $x > A_k$. As a result

$$(z - A_k)^{\beta_k} = \begin{cases} (x - A_k)^{\beta_k} & \text{if } x \text{ is real and } x > A_k, \\ |x - A_k|^{\beta_k} e^{i\pi\beta_k} & \text{if } x \text{ is real and } x < A_k. \end{cases}$$

The complex plane slit along the union of the rays $\cup_{k=1}^n \{A_k + iy : y \leq 0\}$ is simply connected (see Exercise 19), so the integral that defines $S(z)$ is holomorphic in this open set. Since the requirement $\beta_k < 1$ implies that the singularities $(\zeta - A_k)^{-\beta_k}$ are integrable near A_k, the function S is continuous up to the real line, including the points A_k, with $k = 1, \ldots, n$. Finally, this continuity condition implies that the integral can be taken along any path in the complex plane that avoids the union of the *open* slits $\cup_{k=1}^n \{A_k + iy : y < 0\}$.

Now

$$\left| \prod_{k=1}^n (\zeta - A_k)^{-\beta_k} \right| \leq c|\zeta|^{-\sum \beta_k}$$

for $|\zeta|$ large, so the assumption $\sum \beta_k > 1$ guarantees the convergence of the integral (5) at infinity. This fact and Cauchy's theorem imply that $\lim_{r\to\infty} S(re^{i\theta})$ exists and is independent of the angle θ, $0 \leq \theta \leq \pi$. We call this limit a_∞, and we let $a_k = S(A_k)$ for $k = 1, \ldots, n$.

[6]Note that the case $\sum \beta_k \leq 1$, which occurs in Examples 1 and 2 above is excluded. However, a modification of the proposition that follows can be made to take these cases into account; but then $S(z)$ is no longer bounded in the upper half-plane.

Proposition 4.1 *Suppose $S(z)$ is given by* (5).

(i) *If $\sum_{k=1}^n \beta_k = 2$, and $\mathfrak{p}$ denotes the polygon whose vertices are given (in order) by $a_1, \ldots, a_n$, then S maps the real axis onto $\mathfrak{p} - \{a_\infty\}$. The point a_∞ lies on the segment $[a_n, a_1]$ and is the image of the point at infinity. Moreover, the (interior) angle at the vertex a_k is $\alpha_k\pi$ where $\alpha_k = 1 - \beta_k$.*

(ii) *There is a similar conclusion when $1 < \sum_{k=1}^n \beta_k < 2$, except now the image of the extended line is the polygon of $n+1$ sides with vertices $a_1, a_2, \ldots, a_n, a_\infty$. The angle at the vertex a_∞ is $\alpha_\infty\pi$ with $\alpha_\infty = 1 - \beta_\infty$, where $\beta_\infty = 2 - \sum_{k=1}^n \beta_k$.*

Figure 6 illustrates the proposition. The idea of the proof is already captured in Example 1 above.

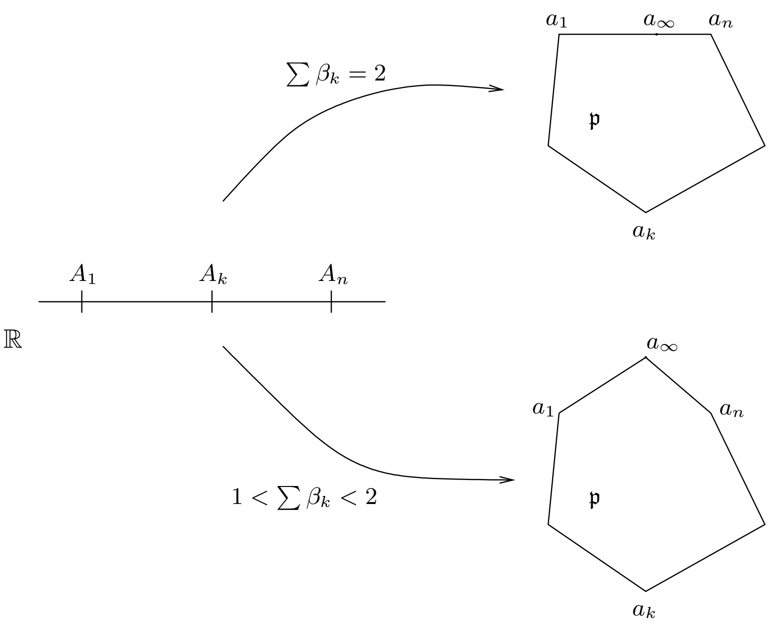

Figure 6. Action of the integral $S(z)$

Proof. We assume that $\sum_{k=1}^n \beta_k = 2$. If $A_k < x < A_{k+1}$ when $1 \le k \le n-1$, then

$$S'(x) = \prod_{j \le k} (x - A_j)^{-\beta_j} \prod_{j > k} (x - A_j)^{-\beta_j}.$$

Hence

$$\arg S'(x) = \arg\left(\prod_{j>k}(x - A_j)^{-\beta_j}\right) = \arg\prod_{j>k} e^{-i\pi\beta_j} = -\pi\sum_{j>k}\beta_j,$$

which of course is constant when x traverses the interval (A_k, A_{k+1}). Since

$$S(x) = S(A_k) + \int_{A_k}^{x} S'(y)\,dy,$$

we see that as x varies from A_k to A_{k+1}, $S(x)$ varies from $S(A_k) = a_k$ to $S(A_{k+1}) = a_{k+1}$ along the straight line segment[7] $[a_k, a_{k+1}]$, and this makes an angle of $-\pi\sum_{j>k}\beta_j$ with the real axis. Similarly, when $A_n < x$ then $S'(x)$ is positive, while if $x < A_1$, the argument of $S'(x)$ is $-\pi\sum_{k=1}^{n}\beta_k = -2\pi$, and so $S'(x)$ is again positive. Thus as x varies on $[A_n, +\infty)$, $S(x)$ varies along a straight line (parallel to the x-axis) between a_n and a_∞; similarly $S(x)$ varies along a straight line (parallel to that axis) between a_∞ and a_1 as x varies in $(-\infty, A_1]$. Moreover, the union of $[a_n, a_\infty)$ and $(a_\infty, a_1]$ is the segment $[a_n, a_1]$ with the point a_∞ removed.

Now the increase of the angle of $[a_{k+1}, a_k]$ over that of $[a_{k-1}, a_k]$ is $\pi\beta_k$, which means that the angle at the vertex a_k is $\pi\alpha_k$. The proof when $1 < \sum_{k=1}^{n}\beta_k < 2$ is similar, and is left to the reader.

As elegant as this proposition is, it does not settle the problem of finding a conformal map from the half-plane to a given region P that is bounded by a polygon. There are two reasons for this.

1. It is not true for general n and generic choices of $A_1, \ldots, A_n$, that the polygon (which is the image of the real axis under S) is simple, that is, it does not cross itself. Nor is it true in general that the mapping S is conformal on the upper half-plane.

2. Neither does the proposition show that starting with a simply connected region P (whose boundary is a polygonal line $\mathfrak{p}$) the mapping S is, for certain choices of $A_1, \ldots, A_n$ and simple modifications, a conformal map from $\mathbb{H}$ to P. That however is the case, and is the result whose proof we now turn to.

[7] We denote the closed straight line segment between two complex numbers z and w by $[z, w]$, that is, $[z, w] = \{(1-t)z + tw : t \in [0, 1]\}$. If we restrict $0 < t < 1$, then (z, w) denotes the open line segment between z and w. Similarly for the half-open segments $[z, w)$ and $(z, w]$ obtained by restricting $0 \le t < 1$ and $0 < t \le 1$, respectively.

4.3 Boundary behavior

In what follows we shall consider a **polygonal region** P, namely a bounded, simply connected open set whose boundary is a polygonal line $\mathfrak{p}$. In this context, we always assume that the polygonal line is closed, and we sometimes refer to $\mathfrak{p}$ as a **polygon**.

To study conformal maps from the half-plane $\mathbb{H}$ to P, we consider first the conformal maps from the disc $\mathbb{D}$ to P, and their boundary behavior.

Theorem 4.2 *If $F : \mathbb{D} \to P$ is a conformal map, then F extends to a continuous bijection from the closure $\overline{\mathbb{D}}$ of the disc to the closure $\overline{P}$ of the polygonal region. In particular, F gives rise to a bijection from the boundary of the disc to the boundary polygon $\mathfrak{p}$.*

The main point consists in showing that if z_0 belongs to the unit circle, then $\lim_{z\to z_0} F(z)$ exists. To prove this, we need a preliminary result, which uses the fact that if $f : U \to f(U)$ is conformal, then

$$\text{Area}(f(U)) = \iint_U |f'(z)|^2 \, dx \, dy.$$

This assertion follows from the definition, $\text{Area}(f(U)) = \iint_{f(U)} dx\,dy$, and the fact that the determinant of the Jacobian in the change of variables $w = f(z)$ is simply $|f'(z)|^2$, an observation we made in equation (4), Section 2.2, Chapter 1.

Lemma 4.3 *For each $0 < r < 1/2$, let C_r denote the circle centered at z_0 of radius r. Suppose that for all sufficiently small r we are given two points z_r and z_r' in the unit disc that also lie on C_r. If we let $\rho(r) = |f(z_r) - f(z_r')|$, then there exists a sequence $\{r_n\}$ of radii that tends to zero, and such that $\lim_{n\to\infty} \rho(r_n) = 0$.*

Proof. If not, there exist $0 < c$ and $0 < R < 1/2$ such that $c \le \rho(r)$ for all $0 < r \le R$. Observe that

$$f(z_r) - f(z_r') = \int_\alpha f'(\zeta)\, d\zeta,$$

where the integral is taken over the arc α on C_r that joins z_r and z_r' in $\mathbb{D}$. If we parametrize this arc by $z_0 + re^{i\theta}$ with $\theta_1(r) \le \theta \le \theta_2(r)$, then

$$\rho(r) \le \int_{\theta_1(r)}^{\theta_2(r)} |f'(z)| r \, d\theta.$$